1.Input-Data capture objectives

In an information system, input is the raw data that is processed to produce output. During the input design, the developers must consider the input devices such as PC, MICR, OMR, etc.

The input design is the link that with information system and user. There are general input concerns that all analysts should have. Input design consists of procedures for data preparation as well data entry.

Here **data procedures** means by those steps necessary to put transaction data into a usable form for processing. And **data entry** mean by activity of putting the data into computer for processing. Data entry can be achieved by computer to read data from a written or printed document, or by having data directly into the system.

It covers all stages from recording of basic data to the feeding of this data into computer for processing the basic steps are

- Original recording
- Data transmission
- Data preparation
- Data verification
- Sorting
- Control
- Computer input

There are several objective of the design of input are,

• **Controlling amount of input:** Because data entry operations are dependent of people, if possible less data entry leads low cost. Second as data input is high leads much time consumption for processing.

• **Avoiding Delay:** This means by, how analyst can design input of the system so that time delay can be reduced for input.

• **Avoiding Errors in Data:** The rate of errors directly depends on data input. The analyst can also affect the error rates of an operation through input design by the way data be entered.

• **Avoiding extra steps:** When the volume of transactions cannot be reduced, the analyst must be sure the process is as efficient as possible. Perfect analyst will also avoid input designs that cause extra steps. This will save time as well less data entry.

• **Keeping the process simple:** the users always accept Simple input design. It is advisable to avoid complexity when there are simple alternatives for

2.Methods for input/capture data

The methods in which the data are captured for transactions are called methods of data capture. Variety of methods is adopted in this concern. Those methods are Source data capture with keypunching, key-to-storage, with scanner and with intelligent terminals. Let us take look to individuals:

Source data capture with keypunching: Best-known and most widely used data entry method is key punching. Logical code is transcribed in punching cards. This verification and correction can be done using computer. A card reader provides a final validation check to ensure against missing data and other errors during preparation and entry. The use of punch cards has largely given way to key-to-storage devices. Best example of this system is employee attendance punch card system. This tracks employee attendance on certain time frame.

Source data capture with key-to-storage: When punch card system fails and times when user require data to be entered like text, one has only alternate like key-to-storage data capture method. Key are alphanumeric values as well special signs inputted from key board.

Contents are always stored on storage devices. Storage devices can be temporary or permanent, primary or secondary. Generally data are entered through a workstation, which includes a typewriter-like keyboard for keying the data and a visual display. Sequence of this procedure is, Writing data on source document, If required code data from source document to form acceptable for processing. Process data on tape or disk validate the data as they processed Store data permanently.

Source data capture with scanner: Optical Character processing is possible with scanner machines. A source document that can be directly used as an input document for scanning by the optical character reader (OCR). When transaction occur, the data are written of marked on the scan able form, sometimes with hand-printed letters and numbers and in other cases by marking special symbols or boxes or represent data. Variety of scanner is available in the market, which provides OCR facility. This sort of instruments is useful in huge data entry job works.

Direct Entry through Intelligent Terminals: Intelligent terminals are special machines, which have connection with computers. These terminals receive data from user and passes to computer. This has very sophistication on work and advance technology has to be used to have these facilities. For example, pen scanner can be used to scan bar code of any product which has information of product like rate, qty etc. Ophthalmic doctors use machine to have eye number of patient, output of the same can generate on computer. Future tends to this emerging technology.

3.Difference between Verification and Validation

Verification	*Validation*
1. **Verification** is a static practice of verifying documents, design, code and program.	1. **Validation** is a dynamic mechanism of validating and testing the actual product.
2. It does not involve executing the code.	2. It always involves executing the code.
3. It is human based checking of documents and files.	*3.* It is computer based execution of program.
4. **Verification** uses methods like inspections, reviews, walkthroughs, and Desk-checking etc.	*4.* **Validation** uses methods like black box (functional) testing, gray box testing, and white box (structural) testing etc.
5. **Verification** is to check whether the software conforms to specifications.	5. **Validation** is to check whether software meets the customer expectations and requirements.
6. It can catch errors that validation cannot catch. It is low level exercise.	*6. It can catch errors that verification cannot catch. It is High Level Exercise.*

7. Target is requirements specification, application and software architecture, high level, complete design, and database design etc.	*7.* Target is actual product-a unit, a module, a bent of integrated modules, and effective final product.
8. **Verification** is done by QA team to ensure that the software is as per the specifications in the SRS document.	*8.* ***Validation*** *is carried out with the involvement of testing team.*
9. It generally comes first-done before validation.	*9.* It generally follows after **verification**.

4.Output design and its objective

The design of output is the most important task of any system. During output design, developers identify the type of outputs needed, and consider the necessary output controls and prototype report layouts.

Objectives of Output Design

The objectives of input design are –

- To develop output design that serves the intended purpose and eliminates the production of unwanted output.
- To develop the output design that meets the end users requirements.
- To deliver the appropriate quantity of output.

- To form the output in appropriate format and direct it to the right person.
- To make the output available on time for making good decisions.

Let us now go through various types of outputs –

External Outputs

Manufacturers create and design external outputs for printers. External outputs enable the system to leave the trigger actions on the part of their recipients or confirm actions to their recipients.

Some of the external outputs are designed as turnaround outputs, which are implemented as a form and re-enter the system as an input.

Internal outputs

Internal outputs are present inside the system, and used by end-users and managers. They support the management in decision making and reporting.

There are three types of reports produced by management information –

- **Detailed Reports** – They contain present information which has almost no filtering or restriction generated to assist management planning and control.
- **Summary Reports** – They contain trends and potential problems which are categorized and summarized that are generated for managers who do not want details.
- **Exception Reports** – They contain exceptions, filtered data to some condition or standard before presenting it to the manager, as information.

5.Fact Finding Techniques(FFT)

Accurate information can be collected with help of certain methods/ techniques. These specific methods for finding information of the system are termed as fact finding techniques. Interview, Questionnaire, Record View and Observations are the different fact finding techniques used by the analyst. The analyst may use more than one technique for investigation.

Interviewing

Systems analyst collects information from individuals or groups by interviewing. The analyst can be formal, legalistic, play politics, or be informal; as the success of an interview depends on the skill of analyst as interviewer.

It can be done in two ways –

- **Unstructured Interview** – The system analyst conducts question-answer session to acquire basic information of the system.
- **Structured Interview** – It has standard questions which user need to respond in

either close (objective) or open (descriptive) format.

Advantages of Interviewing

- This method is frequently the best source of gathering qualitative information.
- It is useful for them, who do not communicate effectively in writing or who may not have the time to complete questionnaire.
- Information can easily be validated and cross checked immediately.
- It can handle the complex subjects.
- It is easy to discover key problem by seeking opinions.
- It bridges the gaps in the areas of misunderstandings and minimizes future problems.

Questionnaires

This method is used by analyst to gather information about various issues of system from large number of persons.

There are two types of questionnaires –

- **Open-ended Questionnaires** – It consists of questions that can be easily and correctly interpreted. They can explore a problem and lead to a specific direction of answer.
- **Closed-ended Questionnaires** – It consists of questions that are used when the systems analyst effectively lists all possible responses, which are mutually exclusive.

Advantages of questionnaires

- It is very effective in surveying interests, attitudes, feelings, and beliefs of users which are not co-located.
- It is useful in situation to know what proportion of a given group approves or disapproves of a particular feature of the proposed system.
- It is useful to determine the overall opinion before giving any specific direction to the system project.
- It is more reliable and provides high confidentiality of honest responses.
- It is appropriate for electing factual information and for statistical data collection which can be emailed and sent by post.

Review of Records, Procedures, and Forms

Review of existing records, procedures, and forms helps to seek insight into a system which describes the current system capabilities, its operations, or activities.

Advantages

- It helps user to gain some knowledge about the organization or operations by themselves before they impose upon others.
- It helps in documenting current operations within short span of time as the

procedure manuals and forms describe the format and functions of present system.

- It can provide a clear understanding about the transactions that are handled in the organization, identifying input for processing, and evaluating performance.
- It can help an analyst to understand the system in terms of the operations that must be supported.
- It describes the problem, its affected parts, and the proposed solution.

Observation

This is a method of gathering information by noticing and observing the people, events, and objects. The analyst visits the organization to observe the working of current system and understands the requirements of the system.

Advantages

- It is a direct method for gleaning information.
- It is useful in situation where authenticity of data collected is in question or when complexity of certain aspects of system prevents clear explanation by end-users.
- It produces more accurate and reliable data.
- It produces all the aspect of documentation that are incomplete and outdated.